LunaLink: An Integrated Four-Subsystem Simulation of a Molniya-Type Earth–Moon Relay

Project X – Spacecraft Design: Fundamentals (SS 2026), Technical University of Munich

Individual project · Python · AI-assisted

1 Introduction

LunaLink is a communications relay on a fixed Molniya-type high-eccentricity orbit (500 × 36,000 km altitude, 63.4° inclination) that links an Ottobrunn ground station to an asset near the Moon. The brief permits modelling a single subsystem; this project instead models all four – EPS, TCS, ADCS and TT&C – on one shared orbit and environment backbone, since their couplings (power, heat, pointing and link geometry) drive the design. The tool is written in transparent Python with a headless evidence generator and an interactive Streamlit dashboard, using SI units internally and favouring visible equations over black boxes. One deliberate consistency choice is noted: the fixed 500 × 36,000 km altitudes give a two-body period of 10.685 h rather than the ≈12 h stated in the brief, so the altitudes are taken as authoritative, the computed period is reported, and 36 h (≈3.4 orbits) is simulated, satisfying the ≥3-orbit rule.

2 Assumptions

Every engineering assumption is listed in the tool's configuration and reproduced in Table 1. Values are chosen from the NASA Small-Spacecraft State-of-the-Art report [1] and standard texts [2–4] and are conservative for a preliminary design.

Area	Assumption	Value / basis
Orbit	Argument of perigee; epoch	270° (northern apogee); 2026-07-06 UTC
EPS	Cell eff. (EOL); array; battery; DoD	0.27; 6.0 m²; 4.5 kWh Li-ion; 40%
EPS	Degradation basis	1 MeV-equiv. fluence from belt model (Sec. 3.2)
TCS	Coatings (mixed)	white / MLI / OSR-FEP faces; ε,α per coating
ADCS	Inertia; initial tumble; B-field	uniform 2.0×1.5×1.0 m box, 500 kg; 10 deg/s; dipole (+IGRF check)
TT&C	X-band; UHF	8.4 GHz, 0.6/3.0 m dishes; 450 MHz, 20/18 dBi
TT&C	Required margin; availability	≥3 dB; ITU-R rain at 99% (p=1%)

3 Results and Discussion

3.1 Orbit and space environment

An Earth-centred inertial state is propagated with two-body gravity plus the J2 zonal harmonic using an adaptive Runge–Kutta integrator, and eclipse, ground-station elevation and Sun/Moon geometry are derived on the same time grid. The 63.4° inclination is not arbitrary: it is the critical inclination at which the J2 secular apsidal drift vanishes, so the apogee stays frozen over the northern hemisphere. The analytic Kozai rate gives $d\omega/dt = 0.0005$ deg/day at 63.4°, against -0.171 deg/day of nodal regression. The propagation was cross-checked independently against Orekit 13.1 (8×8 gravity with luni-solar third bodies) and NASA SPICE/DE440: the period matches exactly, the J2-only trajectory stays within 38 km of the full model over 36 h, and the apsidal drift is 0.004 deg/day at 63.4° versus 0.29 deg/day at 45° – a 66× difference that confirms why this orbit is used [2]. A 60-day luni-solar run yields an inclination station-keeping budget of 3.6 m/s per year (18 m/s over the 5-year life).

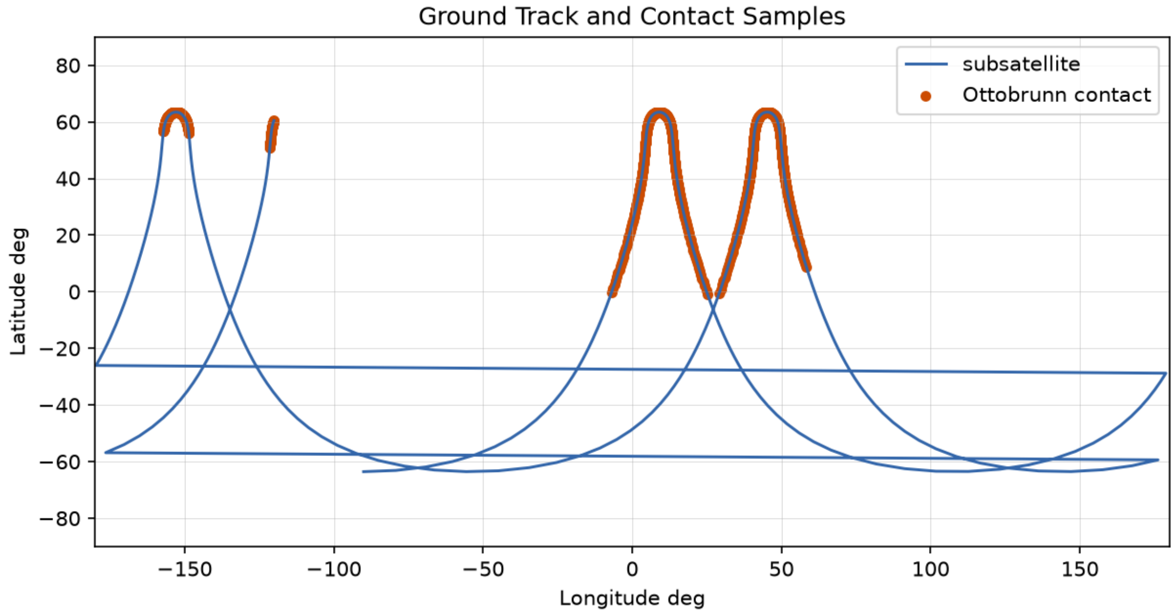

Figure 1. Ground track over 36 h. The apogee dwell over the northern hemisphere gives long Ottobrunn passes; markers show contacts.

3.2 EPS – Electrical Power System

Three power modes are defined (safe, nominal, and peak relay at the 1.2 kW EOL budget) and the array and battery are sized against the worst eclipse. A sun-tracking 6.0 m² array at 27% end-of-life cell efficiency delivers 1927 W, comfortably above the 1.2 kW budget, and a 4.5 kWh Li-ion battery at 40% depth-of-discharge holds the state of charge above 75% with no unserved load (Figure 2). Because a Molniya orbit repeatedly crosses the Van Allen belts, degradation is closed on the loop: mapping the McIlwain L-shell to a trapped-electron flux gives a 1 MeV-equivalent fluence of 5.3e+13 e/cm² over five years and an ionising dose of about 15 krad(Si)/yr behind 2.5 mm of aluminium – in the published 10–30 krad/yr band [1,5]. The derived triple-junction array retains 95% of its power at end of life, which backs the assumed EOL efficiency.

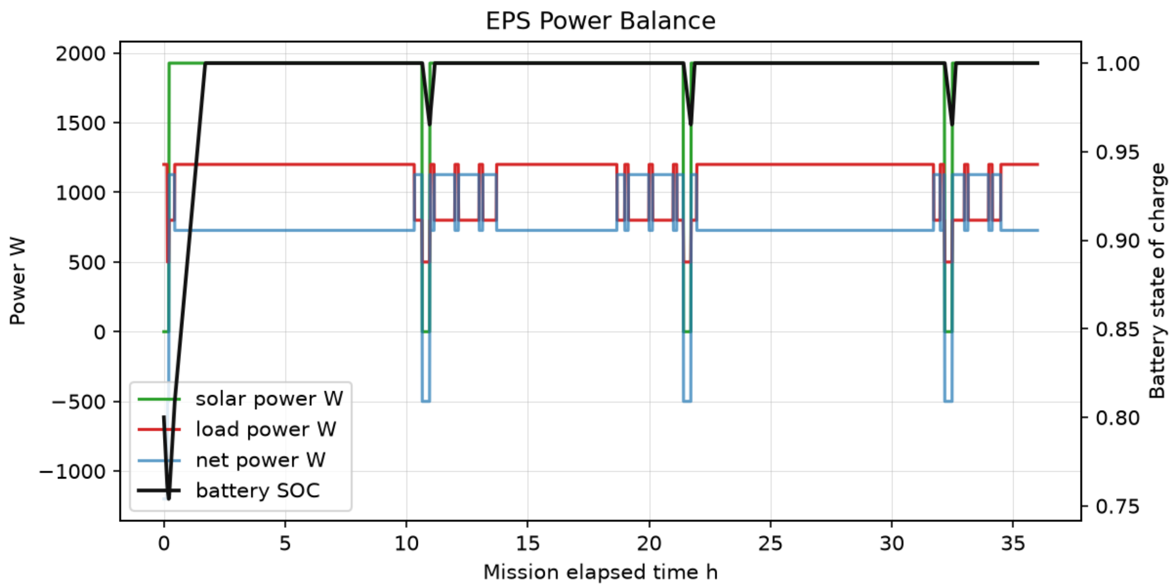

Figure 2. Generation, load and net power (top) and battery state of charge (bottom). Each eclipse drives a discharge that recharges in sunlight.

3.3 TCS – Thermal Control System

The bus is modelled as a seven-node lumped network (six external faces plus one internal equipment node) with radiative exchange to a 3 K sink and the environment fluxes – direct solar, Earth albedo and Earth IR – from the orbit model. Faces carry a mixed coating set (white paint, MLI and an OSR/FEP radiator) chosen to balance the hot and cold cases. Figure 3 shows the external faces swinging with the Sun and eclipse while the internal node, fed by the active dissipation, stays inside the –20/+60 °C electronics band. The worst-case operating margin over the run is 7.6 K and no component limit is violated. This is a lumped-parameter engineering estimate; the face view factors assume an

LVLH-pointing bus and are not coupled to the ADCS quaternion, which is stated as a limitation rather than hidden.

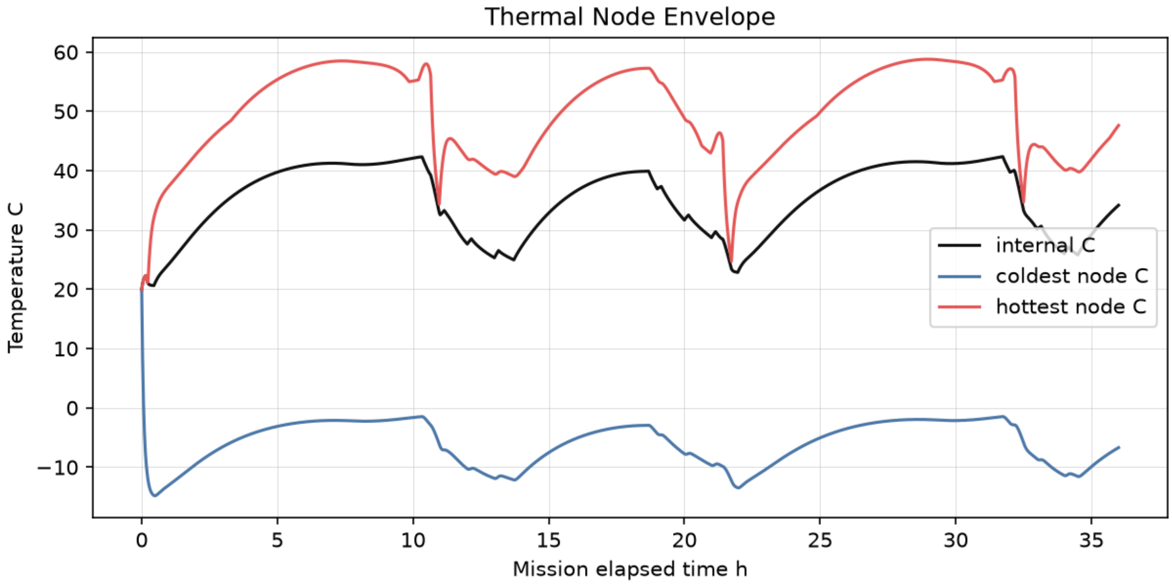

Figure 3. Internal-node and six external-face temperatures over the mission. Faces swing with Sun/eclipse; the internal node stays in limits.

3.4 ADCS – Attitude Determination and Control

The principal inertia is taken from the uniform box and the full Euler rigid-body equations are integrated with a fourth-order Runge–Kutta scheme and quaternion kinematics. A B-dot magnetorquer law detumbles the spacecraft from an initial 10 deg/s to 0.026 deg/s (Figure 4), below the 0.05 deg/s threshold, while the reaction-wheel momentum stays at 0.15 Nms – far from the assumed 12 Nms capacity, so no desaturation is needed. A closed-loop quaternion-feedback PD sun-pointing mode slews the array normal from an initial 67° error to a settled 0.011°, meeting a 3° requirement. The disturbance torques (gravity gradient, solar radiation pressure and residual dipole) are recorded against orbit position. An aligned-dipole field is used for authority sizing and was verified against the full IGRF-14 model (via ppigrf): the two agree to within 5% on average, so the dipole is adequate at this altitude [3,6].

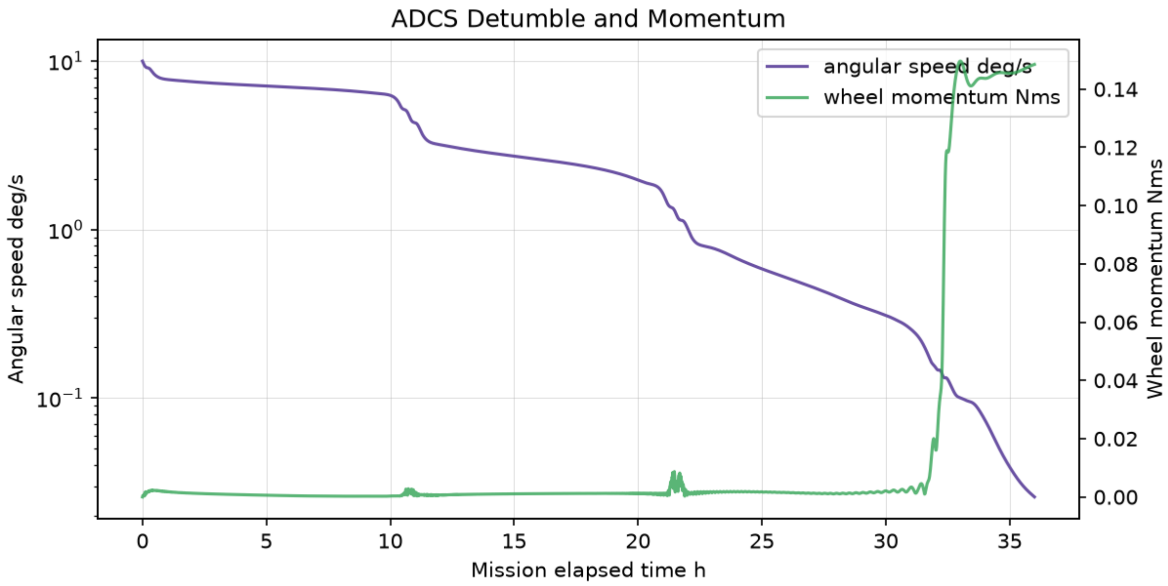

Figure 4. Detumbling convergence, reaction-wheel momentum and disturbance torques. The body rate falls from 10 deg/s to below the 0.05 deg/s threshold.

3.5 TT&C – Telemetry, Tracking and Command

Both links are computed in decibels: EIRP, free-space path loss, G/T, carrier-to-noise density, Eb/N0 and margin. The Earth X-band downlink closes at 100 Mbps with a minimum margin of 5.1 dB during contacts, and the low-rate Moon UHF link keeps a similar reserve, both above the 3 dB requirement (Figure 5). Over 36 h there are four Ottobrunn contact windows totalling about 26 h of visibility – the long apogee dwell that motivates the orbit – and a downlinked volume near 9.4 Tbit. The atmosphere is refined with the ITU-R P.618/P.676 rain-plus-gas model at 99% availability:

the loss rises from 0.21 dB at zenith to 2.57 dB at the 5° window edges, so the low-elevation passes size the margin. CCSDS concatenated coding is specified (about 8 dB of coding gain over uncoded BPSK) and the geometric Doppler is reported, reaching 85 kHz near perigee [7,8].

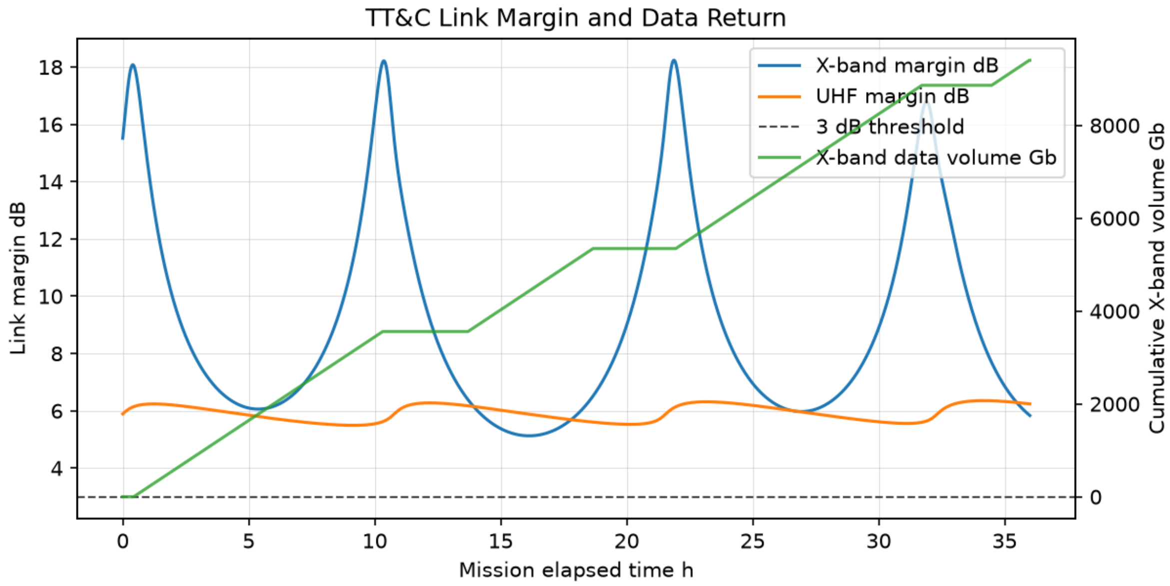

Figure 5. Slant range, link margins and cumulative data volume. The X-band margin stays above the 3 dB requirement during contacts.

Table 2 summarises the two link budgets at their design points.

Quantity	Earth X-band downlink	Moon UHF link
Frequency / rate	8.4 GHz / 100 Mbps	450 MHz / 10 kbps
Required Eb/N0; margin	5 dB; ≥3 dB met	6 dB; ≥3 dB met
Min. margin (contact)	5.1 dB	5.5 dB
Atmosphere (ITU-R)	0.21–2.57 dB (90°–5°)	gaseous, small
Coding (CCSDS)	RS+conv, ~8 dB gain	RS+conv, ~8 dB gain

3.6 Verification

The tool records 26 automated validation checks with pass/warn/fail status; all 26 pass in the baseline run, and the physics is additionally covered by 74 unit tests. Table 3 lists a representative subset spanning every subsystem.

Check	Status	Value
fixed altitude orbit period h	PASS	10.68
orbit frozen apsides argp rate deg per day	PASS	0.0004655
eps minimum state of charge	PASS	0.7539
eps array eol power w	PASS	1927
thermal worst operating margin k	PASS	7.633
adcs final angular speed deg s	PASS	0.0259
adcs sun pointing settled error deg	PASS	0.01122
ttc xband min margin db	PASS	5.127
ttc xband atmos loss 5deg db	PASS	2.574
radiation annual dose krad	PASS	15.25

4 How I Used AI

I used an AI assistant (Anthropic Claude) throughout: to help plan the architecture, to draft and refactor the Python modules and the dashboard, and to speed up documentation. I treated its output as a first draft to be checked, not as truth. Concretely, I independently verified the orbit against Orekit and SPICE, checked every subsystem result against published orders of magnitude and 74 unit tests, and corrected several AI mistakes – for example an incorrect radiation dose constant (off by five orders of magnitude until I anchored it to SHIELDOSE-2-class values), a mixed-dtype table that broke the dashboard cache, and the handling of the brief's period inconsistency. The judgement of what to model, which assumptions to make, and whether each number was physically sensible is my own.

5 References

[1] NASA Ames, *Small Spacecraft Technology State-of-the-Art*, NASA/TP-2022, 2024.

[2] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed., Microcosm/Springer, 2013.

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[4] D. G. Gilmore (ed.), *Spacecraft Thermal Control Handbook*, Vol. 1, Aerospace Press, 2002.

[5] J. I. Vette, *The AE-8 Trapped Electron Model Environment*, NSSDC/WDC-A-R&S 91-24, NASA GSFC, 1991.

[6] P. Alken et al., “International Geomagnetic Reference Field: the thirteenth generation,” *Earth, Planets and Space*, 73:49, 2021.

[7] ITU-R, *Recommendation P.618: Propagation data and prediction methods for Earth-space telecommunication systems*, ITU, 2023.

[8] CCSDS, *TM Synchronization and Channel Coding*, CCSDS 131.0-B-4, 2023.

[9] F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, Springer, 2014.

[10] L. Maisonobe et al., *Orekit: An accurate and efficient core layer for space flight dynamics applications*, orekit.org, 2024.

[11] C. H. Acton, “Ancillary data services of NASA's Navigation and Ancillary Information Facility (SPICE),” *Planetary and Space Science*, 44(1), 1996.